

Liangyu WU | Curriculum Vitae

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Education Background

Stanford University <i>Ph.D. Student in Physics</i> (Advisor: Dr. Julia Gonski & Dr. Dong Su) • Research area: Experimental Particle Physics	California, U.S. Sep. 2024-Present
Shanghai Jiao Tong University <i>Bachelor of Science in Physics, School of Physics and Astronomy</i> • Overall GPA: 86.1/100 (A-); Physics-related GPA: 90.1/100 (A) • Coursework: <i>Quantum Mechanics, Electrodynamics, Equations of Mathematical Physics, Professional Experiment, Introduction to Nuclear and Particle Physics, Methods of Experimental Nuclear and Particle Physics</i> (All graded A's)	Shanghai, China Sep. 2020- Jun. 2024
University of Maryland, College Park <i>Visiting International Student</i> • Overall GPA: 3.925/4.0	Maryland, U.S. Aug. 2023-Dec. 2023

Research Experience

SLAC National Accelerator Laboratory <i>Particle Physics and Astrophysics</i> <i>Research Assistant</i> (Supervisor: Dr. Julia Gonski & Dr. Ariel Schwartzman & Dr. Dong Su) – Research on the ATLAS Experiment at CERN • Foundation model development with ATLAS Run 2 proton-proton collision data. • Advancing LAr timing for primary vertex t_0 reconstruction towards 5D calorimetry with high precision timing. • GigaBit Cable Receiver (GBCR) ASIC QC&SEU testing for the HL-LHC ITk upgrade. – Research on the Future Colliders • On-chip machine learning algorithms in future advanced detectors (dual-readout calorimeters, drift chambers, etc.). • R&D for embedded FPGA (eFPGA) technology in advanced subsystems for future colliders. • Exploration of R&D on Digital SiPM (dSiPM) technology for future detectors.	California, U.S. Jan. 2025-Present
SLAC National Accelerator Laboratory <i>Particle Physics and Astrophysics</i> <i>Research Assistant</i> (Supervisor: Prof. Spencer Gessner) – Research on the FACET-II • Enhancements to the X-Band TCAV GUI and associated analysis tools.	California, U.S. Sep. 2024-Dec. 2024
University of Maryland, College Park <i>Department of Physics</i> <i>Undergraduate Research Student</i> (Supervisor: Prof. Christopher Palmer & Prof. Sarah Eno) – Research on the Dual-Readout Calorimetry • Investigation of single-particle responses through Geant4 simulations across diverse calorimeters. • Formula derivation which predicts the dual-readout-corrected energy from scintillator and Cherenkov signals alone.	Maryland, U.S. Aug. 2023-Aug. 2024
Shanghai Jiao Tong University <i>School of Physics and Astronomy</i> <i>Undergraduate Research Assistant</i> (Supervisor: Prof. Yue Meng) – R&D for a Novel Radon Detector • Designed and constructed the Radic detector and determined the diffusion coefficient of various membrane materials.	Shanghai, China Oct. 2021-Feb. 2024

- **Work in Ultra-low background technique R&D group**
- Measured radon emanation rates across diverse materials for the PandaX and JUNO experiments.
 - Conducted ultra-low radioactive surface treatments of materials for TPC assembly in PandaX-4T experiment.
- **Work in PMT group**
- Performed PMT high-voltage testing to ensure standard operating currents and assessed parameters.
 - Participated in the installation of veto PMTs and high-voltage cables for TPC-bottom PMTs.

- **Research on the stability of unevenly spaced planetary systems**
- Conducted N -body simulations to study the planetary system stability.
 - Demonstrated that using evenly spaced models can overestimate orbital stability time in natural systems.

Publications

[1] D. Yilmaz, **L. Wu**, J. Gonski, Edge Machine Learning for Cluster Counting in Next-Generation Drift Chambers, [**Accepted by NeurIPS 2025**]

[2] S. Eno, **L. Wu** *et al.*, On the resolution of dual readout calorimeters, **NIM A**, **Volume 1083**, **2026**, **171080**, **ISSN 0168-9002**.

[3] **L. Wu**, L. Si *et al.*, Design and Experimental Application of a Radon Diffusion Chamber for Determining Diffusion Coefficients in Membrane Materials, **JINST**, **20**, **no.03**, **P03031 (2025)**

[4] S.V. Chekanov, S. Eno, S. Magill, C. Palmer, **L. Wu**, Geant4 simulations of sampling and homogeneous hadronic calorimeters with dual readout for future colliders, **NIM A**, **Volume 1072**, **2025**, **170200**, **ISSN 0168-9002**.

[5] S. Yang, **L. Wu** *et al.*, The stability of unevenly spaced planetary systems, **ICARUS**, **Volume 406**, **December 2023**, **115757**

Conference Talks

1. [Oct 2025] **Coordinating Panel on Advanced Detectors (CPAD) 2025**
- *Developing Digital Silicon Photomultiplier (dSiPM) Specifications for a High-Granularity Calorimeter with Simulations*
 - *Embedded ML Solutions for Real-time Processing in Future Drift Chambers*
2. [Jul 2025] **US ATLAS Summer Workshop 2025**
- *Enhancing Timing Information Reconstruction with LAr Calorimeter*
3. [May 2025] **ITk Week 2025**
- *Recent Results of GBCR3 Testing at SLAC*

Skills

Technical: C++, Python, CERN ROOT, Geant4, SolidWorks, COMSOL, LaTeX

Hardware: Ultra-low radioactive technology, High-voltage technology, PCB debugging

Language: English (Fluent), Mandarin (Native)

Extra-Curricular Experience

Selected Honors and Awards

- 2024 Shanghai Outstanding Graduate honors, Shanghai Education Commission
- 2023 Best Report First Prize, TDLI Astro-Division Winter Camp (5 out of 96 participants)
- 2022 NetClass Education and Research Fund Scholarship: First Prize, Shanghai Jiao Tong University (Top 5%)
- 2021 and 2022 Academic Excellent Scholarship, Shanghai Jiao Tong University (Top 10%, twice)
- 2021 Chun Geng Education Scholarship, Shanghai Jiao Tong University, School of Physics and Astronomy (Top 5%)